

# Evaluating Multilingual Reliability of Large Language Models in Scientific Literature Analysis

## Abstract

Large Language Models (LLMs) have transformed natural language processing across diverse academic domains, serving as core computational engines for scientific literature mining, automated hypothesis generation, biomedical question answering, and evidence-based claim verification. However, their global utility is fundamentally constrained by an acute linguistic disparity: while state-of-the-art LLMs demonstrate high proficiency in English-centric scientific tasks, their operational reliability degrades precipitously across non-English and low-resource languages. This paper presents an exhaustive evaluation of the multilingual reliability of LLMs within scientific literature analysis. We examine the structural, tokenological, and semantic mechanisms driving cross-lingual performance degradation. By analyzing recent benchmarks—including the Confabulations from ACL Publications (CAP) dataset, the Indic-QA benchmark, BenchMAX, mSTEB, and PubMedQA—we quantify disparities in factual accuracy, semantic consistency, and judgment stability across diverse language families. We critically evaluate the "LLM-as-a-Judge" evaluation paradigm, demonstrating that model scaling and multilingual fine-tuning fail to yield cross-lingual judgment consistency ( $\kappa \approx 0.3$ ).

Furthermore, we analyze the Translate-Test paradigm, open-versus-proprietary architecture performance gaps, and token fertility biases that impose compute and context penalties on morphologically complex languages. Finally, we establish a tactical research roadmap to foster equitable, cross-lingual scientific NLP systems.

**Keywords:** Multilingual Large Language Models, Scientific Literature Analysis, Cross-Lingual Consistency, Scientific Hallucination, Token Fertility Bias, LLM-as-a-Judge.

## Introduction

The exponential growth of peer-reviewed scientific publications presents a monumental challenge for knowledge synthesis, evidence-based policy making, and interdisciplinary research. Large Language Models have emerged as viable tools capable of parsing, summarizing, and reasoning over vast scientific corpora. Specialized architectures and instruction-tuned variants have achieved impressive results on benchmark datasets such as PubMedQA for biomedical reasoning and SciQAG for multi-domain scientific question answering. These capabilities offer the potential to democratize access to scientific discovery and accelerate systematic literature reviews.

Despite these advancements, the global scientific ecosystem operates within a linguistically diverse reality. While English dominates indexed scientific publications, substantial scientific discourse, regional clinical trials, agricultural studies, and environmental assessments are produced in non-English languages. Existing LLMs remain heavily English-centric due to

deep-seated biases in pre-training data composition. When deployed to analyze scientific literature in non-English languages—particularly low-resource or morphologically complex languages—LLMs demonstrate fragile reasoning, elevated hallucination rates, and profound semantic inconsistencies.

Evaluating the multilingual reliability of LLMs in scientific contexts requires looking beyond general-domain conversational fluency. Scientific text is characterized by precise technical terminology, dense statistical arguments, domain-specific ontology mappings, and strict factual bounds. A minor translation error or subword tokenization distortion can fundamentally alter the interpretation of p-values, clinical trial outcomes, or chemical synthesis protocols. Moreover, emerging meta-evaluation strategies—such as using LLMs as automated evaluators ("LLM-as-a-Judge")—exhibit severe cross-lingual instability, rendering automated evaluation pipelines unreliable across different languages.

This research paper provides a synthesis-driven investigation into the multilingual reliability of LLMs in scientific literature analysis. We analyze the theoretical taxonomy of model consistency, examine empirical performance across recent scientific benchmarks, evaluate the tokenological mechanisms underlying cross-lingual degradation, and assess current mitigation strategies such as the Translate-Test paradigm.

## Background and Related Work

### Evolution of Multilingual Scientific NLP

The landscape of multilingual scientific Natural Language Processing has evolved from domain-specific encoder models to massive generative Multilingual Large Language Models (MLLMs). Early breakthroughs relied on masked language modeling fine-tuned on specialized scientific corpora. For example, BioBERT and BioELECTRA established strong baselines for biomedical question answering by pre-training on PubMed abstracts and PMC full-text articles. Concurrently, cross-lingual encoder models such as XLM-RoBERTa demonstrated that unsupervised representation learning across broad multilingual corpora could facilitate cross-lingual transfer.

The paradigm shifted with generative auto-regressive LLMs, including proprietary models (such as GPT-4o and Gemini 2.0 Flash) and open-weight architectures (such as Qwen 2 Audio, Gemma 3, and DeepSeek-V3). While these models demonstrate zero-shot and in-context learning capabilities, their pre-training distributions remain overwhelmingly skewed toward high-resource languages, predominantly English and Chinese.

### Scientific Literature QA and Retrieval Benchmarks

To measure model performance in scientific reasoning, various structured datasets have been established:

- **PubMedQA:** Introduced by Jin et al. (2019), PubMedQA evaluates biomedical reasoning over PubMed abstracts requiring yes/no/maybe responses based on quantitative and clinical evidence.
- **SciQAG & SciQAG-24D:** Developed to expand scientific question answering across 24 distinct scientific domains, leveraging auto-generated and expert-validated paper-question pairs.

- **Indic-QA**: A context-grounded benchmark covering 11 Indian languages across extractive and abstractive question answering tasks, highlighting severe performance drops in low-resource settings.
- **CAP (Confabulations from ACL Publications)**: A dedicated dataset designed by Gamba et al. (2025) to probe scientific hallucinations and cross-lingual factuality versus fluency across nine languages using computational linguistics literature<sup>1</sup>.
- **mSTEB**: A massively multilingual benchmark spanning over 100 speech languages and 200 text languages to evaluate multi-task capabilities, including reading comprehension, translation, and topic classification<sup>3</sup>.

The broader LLM evaluation spectrum is structured around three primary pillars: domain capability evaluation, alignment and consistency measurement, and safety and reliability verification.

## Main Thematic and Technical Discussion

### Dimensionalizing Reliability: Semantic, Logical, and Cross-Lingual Consistency

Assessing the reliability of LLMs in scientific processing requires defining consistency metrics. Reliability cannot be reduced to simple accuracy scores; it demands invariance under semantically preserving transformations across languages.

Logical consistency measures an LLM's adherence to formal logical principles. Negational

consistency mandates that for a scientific statement  $p$ , if  $p \iff \neg(\neg p)$ , the model's

predictions for  $p$  and  $\neg p$  must be mutually exclusive. Symmetric consistency requires that for

a binary relation  $f(x, y)$ , swapping input order must preserve logical truth where symmetric equivalence holds, defined mathematically as:

$$f(x, y) = f(y, x)$$

Transitive consistency evaluates deductive natural language inference. If a model predicts that

$X \rightarrow Y$  and  $Y \rightarrow Z$ , it must deductively yield  $X \rightarrow Z$ .

Semantic consistency requires an LLM to generate invariant outputs across semantically equivalent contexts:

$$f(X) = f(Y) \quad \text{if } X \equiv Y$$

In multilingual scientific literature analysis, semantic consistency dictates that a query posed in

English ( $X_{\text{en}}$ ) and its technical translation in French ( $X_{\text{fr}}$ ) or Malayalam ( $X_{\text{ml}}$ ) should

extract identical factual conclusions from a scientific text. However, empirical studies confirm that LLMs frequently violate semantic consistency across language boundaries, offering conflicting clinical or empirical interpretations depending on the prompt's language. Cross-lingual judgment consistency measures inter-language agreement when LLMs grade or summarize scientific outputs. This is quantified using Fleiss' Kappa ( $\kappa$ ):

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

where  $\bar{P}$  represents the observed mean agreement across linguistic variants of the prompt, and  $\bar{P}_e$  is the expected agreement by chance. Recent benchmarks demonstrate an average Fleiss' Kappa of approximately 0.3 across 25 languages, indicating poor inter-lingual judgment reliability.

### Tokenization Inefficiency and Token Fertility Bias

A core technical driver of multilingual unreliability in scientific literature analysis is tokenization inefficiency. Standard LLM tokenizers, such as Byte-Pair Encoding or WordPiece, are optimized on English-dominated corpora. Consequently, non-English text—especially in morphologically rich or agglutinative languages—is fragmented into excessive subword tokens.

The Token Fertility Ratio ( $\mathcal{F}_L$ ) for a given language  $L$  relative to a baseline reference corpus (typically English) is defined as:

$$\mathcal{F}_L = \frac{N_{\text{tokens}}(S_L)}{N_{\text{words}}(S_L)}$$

where  $N_{\text{tokens}}(S_L)$  is the number of generated tokens for a sentence  $S$  translated into language  $L$ , and  $N_{\text{words}}(S_L)$  is the corresponding word count. Analytic languages such as English and French exhibit low token fertility ratios, whereas agglutinative languages such as Telugu, Bengali, or African languages exhibit high fertility ratios. In scientific literature processing, token fertility bias induces three primary systemic failure modes:

1. **Contextual Compression Penalty:** Elevated token fertility exhausts the model's effective context window ( $N_{\text{ctx}}$ ) rapidly. A scientific paper abstract requiring 250 tokens in English may consume over 800 tokens in Gujarati or Telugu, restricting the model's ability to ingest long-context literature.

- 2. **Attentional Dispersion and Noise:** Over-fragmentation of technical domain terms (e.g., splitting a complex chemical compound name or medical term into a long sequence of sub-tokens) disrupts self-attention mechanics. The model struggles to maintain token-level co-reference chains across long scientific arguments.
- 3. **Inference Compute Disparity:** High token fertility inflates the computational cost ( $O(N^2)$  quadratic attention complexity) for processing non-English scientific texts, creating economic barriers for non-English scientific institutions.

Scientific Confabulations and Hallucination Dynamics

Scientific hallucination, or confabulation, occurs when an LLM generates plausible-sounding but factually incorrect, unverified, or nonsensical claims regarding scientific findings. In general domain NLP, hallucinations are often benign or easily detectable. In scientific literature analysis, confabulations corrupt domain knowledge by misrepresenting statistical significance, falsifying bibliographic citations, or altering experimental findings.

The CAP dataset provides a framework for analyzing cross-lingual scientific hallucinations<sup>1</sup>. CAP disentangles generation errors into two distinct orthogonal axes: factuality errors (inaccuracies in scientific claims, methodologies, or data interpretations) and fluency errors (syntactic, grammatical, or morphological flaws in the output text).

A critical insight highlighted by CAP is the fluency-factuality decoupling phenomenon: as LLMs scale in parameters, their linguistic fluency in non-English languages improves significantly faster than their cross-lingual scientific factuality<sup>1</sup>. Consequently, an LLM evaluating a scientific paper in Bengali or Italian may generate syntactically immaculate prose that contains catastrophic factual misrepresentations of the underlying research<sup>1</sup>.

An analysis of token-level output logit probabilities from CAP reveals that scientific confabulation rates are strongly correlated with context complexity and question phrasing, rather than external proxy metrics such as paper citation counts<sup>1</sup>. When scientific queries require multi-step reasoning over long-context inputs, logit distributions exhibit high entropy, leading the model to default to surface-level lexical generalizations<sup>1</sup>.

Current Approaches and Benchmark Developments

Comparative Analysis of Leading Scientific and Multilingual Benchmarks

To systematically map the landscape of multilingual scientific evaluation, the following table synthesizes the scope, modality, language coverage, and core focus of dominant research benchmarks.

| Benchmark Dataset | Primary Modality and Domain | Language Coverage | Primary Evaluation Task and Metrics | Core Reliability Finding |
|-------------------|-----------------------------|-------------------|-------------------------------------|--------------------------|
|-------------------|-----------------------------|-------------------|-------------------------------------|--------------------------|

|                              |                                          |                                                      |                                                                   |                                                                                                                             |
|------------------------------|------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>CAP</b><br>[cite: 1, 2]   | Text / Scientific NLP (ACL Publications) | 9 languages (5 High-Resource , 4 Low-Resource Indic) | Factuality and Fluency<br>Hallucination Annotations, Token Logits | Demonstrates decoupling of fluency and factuality; reveals elevated hallucination in low-resource science QA <sup>1</sup> . |
| <b>SciQAG-24D</b>            | Text / 24 Scientific Domains             | English (Monolingual baseline)                       | Open-ended Science QA, Precision, Reasoning Accuracy              | Highlights that fine-tuning improves domain QA, but reveals a lack of broad multilingual coverage.                          |
| <b>Indic-QA</b>              | Text / Context-Grounded QA               | 11 Major Indian Languages                            | Extractive & Abstractive QA, BLEU/ROUGE                           | Uncovers a strong English bias; shows Translate-Test outperforms direct native generation.                                  |
| <b>BenchMAX</b><br>[cite: 4] | Text / Multi-capability Evaluation       | 16 Languages (Machine Translated + Post-Edited)      | Multi-way Cross-Lingual Alignment, Capabilities Selection         | Identifies persistent high-resource bias; shows models like DeepSeek-V3 narrow the proprietary gap <sup>4</sup> .           |
| <b>mSTEB</b><br>[cite: 3]    | Speech & Text / Multi-task               | 100+ Speech, 200+ Text Languages                     | QA, Classification, S2TT, ASR, NLI                                | Identifies massive gaps between proprietary and                                                                             |

|                 |                            |                                                  |                                                     |                                                                                                    |
|-----------------|----------------------------|--------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|
|                 |                            |                                                  |                                                     | open models, with severe degradation in low-resource regions <sup>3</sup> .                        |
| <b>PubMedQA</b> | Text / Biomedical Research | English (Used as source for translation studies) | Closed-form Reasoning (Yes/No/Maybe Classification) | Achieves 96%+ accuracy via ensemble methods in English, but exhibits severe drops when translated. |

## Multilingual Scientific Question Answering and Retrieval

Scientific question answering requires models to parse dense research abstracts and extract precise evidence. Benchmarks such as PubMedQA test whether an LLM can synthesize clinical evidence to answer research questions. On English PubMedQA, state-of-the-art ensemble techniques—such as boosting-based weighted majority voting and dynamic cluster selection—achieve accuracy scores exceeding 96%. However, when these biomedical queries are presented in non-English languages, accuracy degrades rapidly.

In open-domain and dense retrieval settings, datasets such as WebFAQ, MIRACL, and MLDR evaluate cross-lingual document passage retrieval. When retrieval-augmented generation (RAG) pipelines ingest non-English scientific queries, dense retrievers frequently fail to align technical concepts across vector spaces, fetching irrelevant context and causing downstream generator hallucination.

## The CAP Dataset and the SHROOM-CAP Benchmark

The release of the CAP dataset represents an advancement in measuring scientific confabulation<sup>1</sup>. Spanning 900 curated scientific questions grounded in ACL Anthology publications and over 7,000 annotated LLM responses across 16 models, CAP provides token-level logits and binary error labels<sup>1</sup>. CAP covers five high-resource languages (English, French, Hindi, Italian, Spanish) and four low-resource Indic languages (Bengali, Gujarati, Malayalam, Telugu)<sup>1</sup>.

The dataset powered the SHROOM-CAP shared task as part of the CHOMPS workshop, challenging detection frameworks to spot scientific overgeneration and factual errors in cross-lingual settings<sup>5</sup>. Results from SHROOM-CAP confirm that standard reference-free hallucination detectors perform poorly when applied to low-resource scientific outputs, frequently misclassifying fluent hallucinations as ground truth<sup>5</sup>.

## Reliability of Multilingual "LLM-as-a-Judge"

To bypass expensive human annotation, modern NLP relies heavily on advanced LLMs (such

as GPT-4o) as automated judges to evaluate generated text. However, systematic studies into multilingual LLM-as-a-Judge frameworks expose profound reliability deficits.

When evaluating models across 25 languages on judgment tasks, LLMs exhibit an average inter-lingual Fleiss' Kappa score of approximately 0.3. This demonstrates that an LLM judge evaluating a scientific answer in English will assign a vastly different score or rationale when evaluating the exact same semantic answer presented in Swahili, Indic, or South American languages. Empirical data shows that neither scaling parameter count nor expanding multilingual pre-training data resolves this judgment variance.

## Research Challenges and Limitations

### The Translate-Test Dilemma vs. Native Processing

A central debate in multilingual NLP revolves around whether to process queries natively or employ the Translate-Test paradigm. In Translate-Test, non-English scientific queries are translated into English via machine translation (MT), processed by an English-optimized LLM, and the resulting response is translated back into the source language.

In low-resource scientific evaluation benchmarks such as Indic-QA, the Translate-Test paradigm consistently outperforms direct native instruction-tuned LLM generation. This outcome highlights a structural limitation: modern multilingual LLMs fail to internalize language-agnostic scientific concepts. Instead, their deep reasoning capabilities remain locked within English-centric latent sub-spaces. However, Translate-Test introduces failure modes:

- **Contextual Distortion:** Scientific nomenclature and chemical formulas can be corrupted by translation engines.
- **Visual-Textual Misalignment:** In multimodal settings (such as parsing figures, charts, and diagrams in scientific papers), translation engines disregard visual text, creating misalignment between translated questions and original figures.

### Proprietary vs. Open-Weight Disparity

Evaluation on benchmarks such as mSTEB and BenchMAX reveals a performance gap between leading proprietary models (e.g., GPT-4o, Gemini 2.0 Flash) and open-weight models (e.g., Qwen 2 Audio, Gemma 3)<sup>3</sup>. In text-based scientific reasoning, advanced open models like Gemma 3 27B and DeepSeek-V3 demonstrate competitive performance against proprietary baselines in high-resource languages<sup>4</sup>. However, in speech-based scientific processing and low-resource text tasks, open-weight models exhibit significant performance degradation<sup>3</sup>.

### Machine Translation Noise and Data Contamination

Many multilingual evaluation benchmarks rely on machine-translated task data derived from English datasets. Automated machine translation introduces subtle grammatical biases, artifact leakage, and lost domain nuances. BenchMAX demonstrates that machine-translated task benchmarks often yield inconsistent capability assessments compared to datasets manually post-edited by native-speaking domain experts<sup>4</sup>.

Furthermore, data contamination remains a critical risk in scientific LLM evaluation. Because frontier LLMs scrape vast portions of the open web, public scientific repositories (e.g., arXiv, PubMed) may be memorized during pre-training. High benchmark accuracy may reflect memorization of English abstracts rather than genuine cross-lingual scientific reasoning.

# Research Gaps and Future Directions

To systematically address the current limitations in multilingual scientific literature analysis, future research should focus on four tactical pillars:

1. **Dynamic Tokenization and Adaptive Subword Vocabularies:** To eliminate token fertility bias, future LLM architectures must move beyond static, English-skewed tokenizers. Research into dynamic, language-agnostic subword tokenization, morphologically aware token splitters, and direct byte-level language modeling is essential to ensure compute parity for non-English scientific literature.
2. **Mechanistic Interpretability of Cross-Lingual Scientific Knowledge:** Investigating how scientific factual knowledge is represented across language boundaries inside an LLM's transformer layers remains critical. Determining whether factual scientific knowledge is stored in language-neutral conceptual hubs or fragmented language-specific circuits will enable targeted model editing and alignment. Advanced probing techniques should focus on tracing how technical terms alter attention weights across high- and low-resource prompt variants.
3. **Culturally and Regionally Grounded Scientific Knowledge Graphs:** Existing scientific QA datasets are overwhelmingly derived from Western academic databases. Constructing native multilingual scientific benchmarks sourced directly from non-Western literature, regional clinical trial registers, and local agricultural publications is vital. Integrating multilingual Knowledge Graphs into Retrieval-Augmented Generation frameworks can ground LLM inference in verified domain facts, mitigating cross-lingual hallucinations.
4. **Standardized Multilingual Meta-Evaluation Protocols:** Given the demonstrated instability of LLM-as-a-Judge approaches ( $\kappa \approx 0.3$ ), standardized evaluation protocols are necessary. Future frameworks should incorporate multi-prompt ensemble voting, semantic consistency verification checks, and cross-lingual calibration constraints before deploying LLMs as automated judges in scientific literature reviews.

## Conclusion

The deployment of Large Language Models for scientific literature analysis offers significant utility, yet its realization is constrained by cross-lingual unreliability. As demonstrated across scientific benchmarks—including CAP, Indic-QA, BenchMAX, mSTEB, and PubMedQA—LLMs suffer from accuracy drops, high confabulation rates, and token fertility penalties when operating in non-English scientific domains<sup>1</sup>.

Increasing model scale and expanding multilingual pre-training data do not automatically resolve fundamental deficits in semantic consistency or meta-evaluation reliability. The decoupling of linguistic fluency from factual accuracy in low-resource scientific text generation presents a subtle risk, as grammatically pristine outputs can mask critical domain errors<sup>1</sup>. Furthermore, the current superiority of the Translate-Test paradigm over native processing underscores that existing LLMs have yet to achieve language-agnostic latent scientific representations.

Bridging this multilingual scientific divide requires targeted innovations: building native non-English scientific benchmarks, designing tokenizers that eliminate fertility bias, grounding

inference in domain knowledge graphs, and formulating mathematically stable cross-lingual evaluation metrics. Advancing these frontiers is essential to build equitable, accurate, and globally reliable scientific language models.

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